AWS CLOUD NETWORKING PROJECT

AWS VPC Peering

Implementation and Verification Report

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Project	AWS VPC Peering
Region	ap-southeast-2 (Sydney)
Report date	31 October 2026
Test verification	0 of 5 tests verified as Passed with evidence

Generated automatically by generate_report.py from project_config.json. Results that are not backed by supplied evidence are marked **Not Verified**.

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1. Executive Summary

This report documents the AWS VPC peering project of Rudra Sanjay Rakshe in the ap-southeast-2 region (Sydney), connecting VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16).

This project connects two isolated Amazon VPCs in the ap-southeast-2 (Sydney) region using a VPC peering connection. A bastion host in the public subnet of VPC A is used to administer private instances in both VPCs.

1.1 Verification Summary

Tests defined	5
Passed (with evidence)	**0**
Failed	0
Not Verified	**5**
Screenshots supplied	0 of 5 listed
Cleanup steps with evidence	0 of 7

Evidence notice: items marked **Not Verified** have no supporting evidence in the supplied configuration. They must not be read as passed or completed.

2. Project Objectives

- Create two isolated VPCs with non-overlapping CIDR blocks in ap-southeast-2.
- Establish a VPC peering connection between VPC A and VPC B.
- Configure route tables so the private subnets can reach each other over the peering connection.
- Restrict access with security groups and a bastion host.
- Verify connectivity with documented tests and screenshot evidence.
- Clean up all resources afterwards to avoid unnecessary charges.

3. Architecture

VPC A (10.10.0.0/16) contains a public subnet with a bastion host and a private subnet with the App-A instance. VPC B (10.20.0.0/16) contains a single private subnet with the App-B instance. A VPC peering connection links the two VPCs.

Peering is not transitive and the CIDR blocks do not overlap. Each VPC route table needs an explicit route to the other VPC's CIDR block through the peering connection.

3.1 VPC Design

VPC	CIDR block	Notes
VPC A	10.10.0.0/16	Hosts bastion and App-A
VPC B	10.20.0.0/16	Hosts App-B

3.2 Subnet Design

Subnet	CIDR block	VPC	Type
Public subnet A	10.10.1.0/24	VPC A	Public
Private subnet A	10.10.2.0/24	VPC A	Private
Private subnet B	10.20.1.0/24	VPC B	Private

3.3 Instances

Instance	Private IP	Subnet	VPC	Role
Bastion host	*Not supplied*	Public subnet A	VPC A	SSH jump host
App-A	10.10.2.10	Private subnet A	VPC A	Application server
App-B	10.20.1.10	Private subnet B	VPC B	Application server

3.4 Diagram and Evidence

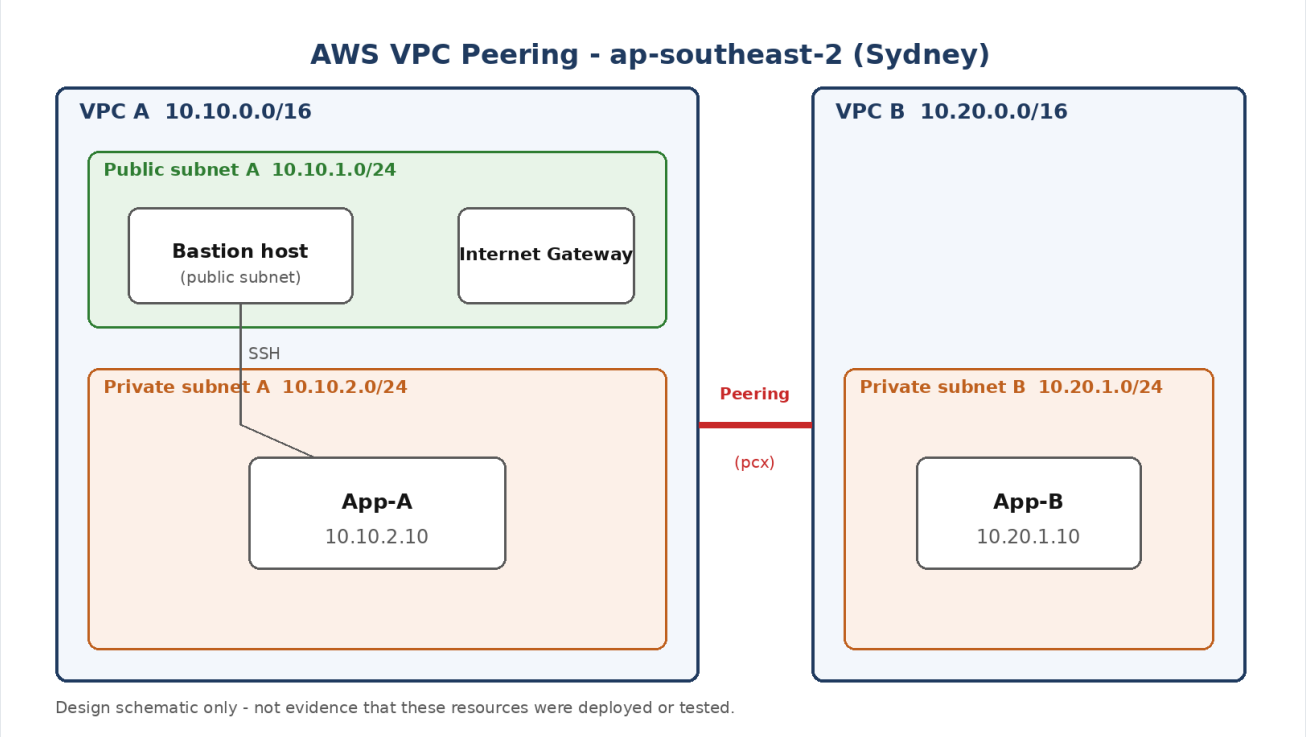


Figure 1: AWS VPC Peering architecture (design schematic, not evidence of deployment)

NOT VERIFIED - screenshot not available
Reason: file not found
Path in config: screenshots/peering_connection_active.png

Figure 2: VPC peering connection in the Active state

4. Resource Inventory

Resource identifiers are reproduced as supplied. An empty identifier means the ID was not recorded; it does not imply the resource does or does not exist.

4.1 VPCs

Name	Resource ID	CIDR / IP address	Details
VPC A	*Not supplied*	10.10.0.0/16	*Not supplied*
VPC B	*Not supplied*	10.20.0.0/16	*Not supplied*

4.2 Subnets

Name	Resource ID	CIDR / IP address	Details
Public subnet A	*Not supplied*	10.10.1.0/24	VPC A
Private subnet A	*Not supplied*	10.10.2.0/24	VPC A
Private subnet B	*Not supplied*	10.20.1.0/24	VPC B

4.3 Gateways and peering

Name	Resource ID	CIDR / IP address	Details
Internet Gateway (VPC A)	*Not supplied*	*Not supplied*	Public subnet A internet access
VPC peering connection	*Not supplied*	*Not supplied*	VPC A <-> VPC B

4.4 EC2 instances

Name	Resource ID	CIDR / IP address	Details
Bastion host	*Not supplied*	*Not supplied*	Public subnet A
App-A	*Not supplied*	10.10.2.10	Private subnet A
App-B	*Not supplied*	10.20.1.10	Private subnet B

5. Routing Configuration

Route entries below are reproduced from the project configuration as written by the author. The generator cannot check them against the live AWS account; see the Test Results section and the screenshots for verification.

5.1 RT-A-Public (VPC A)

Associated subnets: Public subnet A

Destination	Target	Notes
10.10.0.0/16	local	*Not supplied*
0.0.0.0/0	Internet Gateway	Internet access for bastion
10.20.0.0/16	VPC peering connection	Reach VPC B

5.2 RT-A-Private (VPC A)

Associated subnets: Private subnet A

Destination	Target	Notes
10.10.0.0/16	local	*Not supplied*
10.20.0.0/16	VPC peering connection	Reach VPC B

5.3 RT-B-Private (VPC B)

Associated subnets: Private subnet B

Destination	Target	Notes
10.20.0.0/16	local	*Not supplied*
10.10.0.0/16	VPC peering connection	Reach VPC A

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Reason: file not found
Path in config: screenshots/route_table_a_private.png

Figure 3: Route table RT-A-Private with route to 10.20.0.0/16

NOT VERIFIED - screenshot not available
Reason: file not found
Path in config: screenshots/route_table_b_private.png

Figure 4: Route table RT-B-Private with route to 10.10.0.0/16

6. Security Configuration

Security rules are reproduced from the project configuration as supplied by the author and have not been checked against the live AWS account.

6.1 Security group: Bastion-SG

VPC: VPC A **Attached to:** Bastion host

Inbound rules

Protocol	Port	Source	Description
TCP	22	<YOUR-PUBLIC-IP>/32	SSH from administrator (replace placeholder)

Outbound rules

Protocol	Port	Destination	Description
All	All	0.0.0.0/0	Default allow all

6.2 Security group: App-A-SG

VPC: VPC A **Attached to:** App-A

Inbound rules

Protocol	Port	Source	Description
TCP	22	10.10.1.0/24	SSH from bastion subnet
ICMP	All	10.20.0.0/16	Ping from VPC B

Outbound rules

Protocol	Port	Destination	Description
All	All	0.0.0.0/0	Default allow all

6.3 Security group: App-B-SG

VPC: VPC B **Attached to:** App-B

Inbound rules

Protocol	Port	Source	Description
TCP	22	10.10.1.0/24	SSH from bastion subnet via peering
ICMP	All	10.10.0.0/16	Ping from VPC A

Outbound rules

Protocol	Port	Destination	Description
All	All	0.0.0.0/0	Default allow all

6.4 Network ACLs

Name	Subnet(s)	Notes
Default network ACLs	All subnets	Default rules (allow all) left unchanged.

NOT VERIFIED - screenshot not available
Reason: file not found
Path in config: screenshots/sg_app_b.png

Figure 5: App-B security group inbound rules

7. Test Results

5 test(s) defined: 0 Passed (with evidence), 0 Failed, 5 Not Verified. A test is reported as Passed only when a referenced evidence screenshot exists.

Not Verified: 5 test(s) have no result with supporting evidence. Connectivity for those paths has not been demonstrated by this report.

7.1 Summary Table

ID	Test	Path	Expected	Status	Evidence
T1	Bastion to App-A (same VPC)	Bastion host to App-A (10.10.2.10)	0% packet loss	**Not Verified**	*None supplied*
T2	Bastion to App-B (over peering)	Bastion host to App-B (10.20.1.10)	0% packet loss	**Not Verified**	*None supplied*
T3	App-A to App-B (over peering)	App-A (10.10.2.10) to App-B (10.20.1.10)	0% packet loss	**Not Verified**	*None supplied*
T4	App-B to App-A (over peering)	App-B (10.20.1.10) to App-A (10.10.2.10)	0% packet loss	**Not Verified**	*None supplied*
T5	SSH from bastion to App-B	Bastion host to App-B (10.20.1.10)	SSH login succeeds	**Not Verified**	*None supplied*

7.2 Test Details

T1 - Bastion to App-A (same VPC)

Source	Bastion host
Destination	App-A (10.10.2.10)
Command	`ping -c 4 10.10.2.10`
Expected result	0% packet loss
Observed output (as supplied)	*Not supplied*
Status	**Not Verified**
Evidence	*None supplied*

T2 - Bastion to App-B (over peering)

Source	Bastion host
Destination	App-B (10.20.1.10)
Command	`ping -c 4 10.20.1.10`
Expected result	0% packet loss
Observed output (as supplied)	*Not supplied*
Status	**Not Verified**
Evidence	*None supplied*

T3 - App-A to App-B (over peering)

Source	App-A (10.10.2.10)
Destination	App-B (10.20.1.10)
Command	`ping -c 4 10.20.1.10`
Expected result	0% packet loss
Observed output (as supplied)	*Not supplied*
Status	**Not Verified**
Evidence	*None supplied*

T4 - App-B to App-A (over peering)

Source	App-B (10.20.1.10)
Destination	App-A (10.10.2.10)
Command	`ping -c 4 10.10.2.10`
Expected result	0% packet loss
Observed output (as supplied)	*Not supplied*
Status	**Not Verified**
Evidence	*None supplied*

T5 - SSH from bastion to App-B

Source	Bastion host
Destination	App-B (10.20.1.10)
Command	`ssh ec2-user@10.20.1.10`
Expected result	SSH login succeeds
Observed output (as supplied)	*Not supplied*
Status	**Not Verified**
Evidence	*None supplied*

NOT VERIFIED - screenshot not available
Reason: file not found
Path in config: screenshots/ping_app_a_to_b.png

Figure 6: Ping from App-A to App-B

8. Troubleshooting

8.1 Issues Encountered

No issues were recorded by the author. Nothing has been added automatically.

8.2 Reference Checklist

General guidance for VPC peering problems. This is a generic checklist, not a record of problems that occurred in this project.

- Peering request not accepted: the connection must be in the Active state before any traffic can flow.
- Missing route: BOTH VPC route tables need a route to the other VPC's CIDR block that targets the peering connection.
- Route table not associated with the intended subnet (the subnet silently uses the main route table).
- Security group does not allow the source CIDR block of the other VPC for the protocol being tested (for example ICMP for ping, TCP 22 for SSH).
- Network ACL rules blocking inbound or outbound traffic (remember ACLs are stateless).
- Overlapping CIDR blocks cannot be peered; peering is also not transitive.
- SSH key permission or wrong user name when connecting through the bastion host.

9. Lessons Learned

No lessons learned were recorded by the author. Nothing has been added automatically.

10. Cleanup Documentation

Cleanup NOT verified. No cleanup evidence was supplied, so this report does not claim that any AWS resource has been deleted. Resources that still exist can continue to incur charges (for example instances, Elastic IPs and NAT gateways).

Recommended order: delete dependent resources first so that the remaining deletions are not blocked.

#	Action	Resource	Status	Evidence
1	Terminate EC2 instances	Bastion host, App-A, App-B	**Not Verified**	*None supplied*
2	Delete VPC peering connection	VPC A <-> VPC B	**Not Verified**	*None supplied*
3	Delete custom route tables	RT-A-Public, RT-A-Private, RT-B-Private	**Not Verified**	*None supplied*
4	Detach and delete Internet Gateway	Internet Gateway (VPC A)	**Not Verified**	*None supplied*
5	Delete subnets	Public subnet A, Private subnet A, Private subnet B	**Not Verified**	*None supplied*
6	Delete VPCs	VPC A, VPC B	**Not Verified**	*None supplied*
7	Release Elastic IP addresses (if any were allocated)	Elastic IPs	**Not Verified**	*None supplied*

11. Conclusion

0 of 5 defined tests are verified as Passed. 5 are Not Verified and 0 Failed. Only the verified results should be treated as demonstrated.

Appendix A: Evidence Register

Figure	Type	Section	Caption	File (as in config)	Status
Figure 1	Diagram	architecture	AWS VPC Peering architecture (design schematic, not evidence of deployment)	assets/architecture.png	**Provided**
Figure 2	Screenshot	architecture	VPC peering connection in the Active state	screenshots/peering_connection_active.png	**Missing** (file not found)
Figure 3	Screenshot	routing	Route table RT-A-Private with route to 10.20.0.0/16	screenshots/route_table_a_private.png	**Missing** (file not found)
Figure 4	Screenshot	routing	Route table RT-B-Private with route to 10.10.0.0/16	screenshots/route_table_b_private.png	**Missing** (file not found)
Figure 5	Screenshot	security	App-B security group inbound rules	screenshots/sg_app_b.png	**Missing** (file not found)
Figure 6	Screenshot	tests	Ping from App-A to App-B	screenshots/ping_app_a_to_b.png	**Missing** (file not found)

Appendix B: Data Completeness Notes

- Screenshot 'peering-active': file not found (screenshots/peering_connection_active.png) - shown as Not Verified placeholder.
- Screenshot 'route-table-a-private': file not found (screenshots/route_table_a_private.png) - shown as Not Verified placeholder.
- Screenshot 'route-table-b-private': file not found (screenshots/route_table_b_private.png) - shown as Not Verified placeholder.
- Screenshot 'sg-app-b': file not found (screenshots/sg_app_b.png) - shown as Not Verified placeholder.
- Screenshot 'ping-app-a-to-b': file not found (screenshots/ping_app_a_to_b.png) - shown as Not Verified placeholder.